

Supplementary Material

Alignment in Multimodal Large Language Models: A Survey

Author information pending confirmation

S1 SCOPE AND READING GUIDE

This independent supplement expands the mechanism and evaluation comparisons in the survey. It contains analytical distinctions and source locations, not new experiments or a matched model leaderboard. The Modality Gap concerns correspondence and use of task-relevant multimodal evidence; the Intent Gap concerns behavior relative to specified goals and constraints. A method can influence both, and a capability error can remain unattributed. Gap labels describe the intervention’s intended target, not a claim that it closes the gap.

S2 ARCHITECTURE COMPARISON

Table S1 expands the evidence-access attributes compared in the main text. The relevant cost depends on input resolution, context length, trainable components, and inference budget; qualitative trade-offs should not be read as experimentally matched efficiency rankings.

TABLE S1
Architectural interventions and the controlled comparisons needed to evaluate them.

Intervention / examples	Selection mechanism	Assumption to inspect	Controlled comparison
Bridge compression: BLIP-2, TokenPacker [1], [2]	Learned query bottleneck or regional-detail injection	A compact token set retains evidence needed downstream	Match decoder and token budget; test small objects and local attributes separately from global answers.
Discrete interfaces: SpeechT5, AnyGPT [3], [4]	Codebook-based representations and token sequences	Discretization preserves task-relevant information across modalities	Separate codebook reconstruction error from reasoning and sequence-length effects.
Explicit structure: MAIL, VideoTree [5], [6]	Entity-mediated exchange or query-adaptive video hierarchy	Extraction and pruning retain decisive entities or events	Perturb or remove selected and unselected evidence; measure missed-event errors and selection cost.
Spatial/layer access: AGE-VLM, Dense Connector [7], [8]	Spatial supervision or multi-layer feature aggregation	Additional features contribute useful detail rather than redundancy	Match image resolution and compute; isolate feature source, fusion, and spatial supervision.
Modality-indexed experts: VLMO, MoT [9], [10]	Modality-conditioned parameter selection with shared interaction	Specialization preserves cross-modal transfer	Report total and active parameters, per-modality loss, and held-out cross-modal tasks.
Adaptive experts: OneLLM, MoIE [11], [12]	Learned routing among projections or experts	Routing uses task-relevant evidence rather than easy modality cues	Audit assignment distributions and evidence interventions separately; balanced utilization is not evidence reliance.
Graph augmentation: LLaVA-SG, MEAformer [13], [14]	Scene relations or multimodal entity structure	Supplied relations are relevant and sufficiently correct	Compare correct, corrupted, and removed relations; distinguish the MLLM mechanism from entity-alignment foundations.

S3 OPERATIONAL DIAGNOSTIC MATRIX

The contrasts in Table S2 test different hypotheses. Maintain task validity after an edit, hold unrelated inputs fixed, and use a reference appropriate to the edited example. A response change under an arbitrary perturbation is not by itself evidence of grounding.

TABLE S2
Diagnostic contrasts for separating evidence failures, constraint violations, and reasoning-capability errors.

Question	Contrast and reference	Supported interpretation	Boundary / control
Is evidence represented?	Probe or compare representations with known modality correspondence; SAIL and MIR [15], [16].	Evidence availability or cross-modal compatibility.	An informative representation need not be used by the decoder.
Is relevant evidence used?	Change a blue mug to red while keeping the color question fixed; update the correct answer.	Consistent, task-appropriate changes support image dependence.	Include irrelevant-region edits and matched natural examples to detect edit artifacts.
Is a claim supported?	Check object claims against annotations, as in POPE and CHAIR [17], [18].	Unsupported-object response rate.	Does not localize the failure to encoding, fusion, or decoding.
Is the action constraint respected?	Keep the scene fixed and contrast “describe without moving” with an authorized action request.	Violations despite correct perception support an Intent Gap attribution.	Check benign utility; universal refusal is not successful task alignment.
Did reasoning fail after perception?	Verify the read values or objects before testing the subsequent inference; MIRAGE [19].	A capability error can be distinguished from incorrect perception.	Do not force the error into either gap without a further evidence-use or constraint test.
Is the reward judge reliable?	Compare pairwise judgments with human-validated labels in VL-RewardBench and Multimodal RewardBench [20], [21].	Agreement on the benchmark’s categories and candidates.	Audit new policy outputs, presentation order, and style confounders separately.
Do gains transfer?	Re-test evidence and constraint contrasts after domain or modality shifts; include uncertainty and benign-utility checks.	Transfer within the documented target conditions.	Aggregate accuracy can hide opposite movements in grounding and behavioral compliance.

S4 RELATED-SURVEY COVERAGE LOCATIONS

Table S3 supports the main comparison matrix. Dedicated coverage, distributed discussion, and absence of an organizing dimension are different judgments. A cross does not mean that a paper never mentions a topic. Version-specific locations are used where a reviewed preprint differs from its later publication; these are not claims about every version.

TABLE S3
Source locations underlying the related-survey comparison.

Reviewed survey	Coverage locations	Interpretation for the matrix
Caffagni et al. [22], Findings ACL 2024	Sec. 2: model components and training; Sec. 3: capabilities and other modalities; Appendices B–C: training data and evaluation benchmarks.	Objectives are discussed within training, not as a separate objective-family comparison.
Shu et al. [23], Findings EMNLP 2025	Sec. 3: misalignment causes; Sec. 4: mitigation; appendices expand training stages, diagnostics, and benchmarks.	Data, model, training, and evaluation are organizing dimensions; applications are not a dedicated method grouping.
Lu et al. [24], EMNLP 2025	Sec. 3: similarity metrics; Sec. 4: cross-domain evidence; Sec. 5: scale, architectures, objectives, and training/task diversity.	Architecture, objectives, data, training, and transfer applications are discussed, but the organizing focus is representation potential.
Fang et al. [25], publisher PDF	Secs. 3–7: implicit, encoder-level, architectural, behavioral, and inference-time alignment; Sec. 8: synthetic data and risks; Sec. 9: evaluation.	Already covers representation and behavior, inference-time intervention, data risks, and evaluation. Those topics alone are not exclusive contributions of this survey.
Li and Tang [26], arXiv v2, 11 Oct. 2025	Sec. 2.2: datasets; Sec. 3.4: applications; Sec. 4: fusion methods; Sec. 5: challenges with training-module comparisons.	Applications are explicitly covered. Training and evaluation occur within method/challenge comparisons rather than a separate staged-training and diagnostic framework.
Ho et al. [27], publisher PDF	Sec. 4 groups mitigation by data, architecture, inference, loss optimization, and evaluation.	Substantial lifecycle coverage is acknowledged; application domains are not the central organizing structure.
Yu et al. [28], arXiv v2, 23 Mar. 2025	Sec. 3: preference algorithms and applications; Sec. 4: annotation; Sec. 5: evaluation, including rewards and alignment.	Preference methods, supervision, evaluation, and applications are covered; architectural families are not the organizing focus.

S5 REPORTED EVIDENCE AND ITS LIMITS

Table S4 records within-study contrasts supporting the main synthesis. Values are transcribed from the cited papers, not remeasured. Each row preserves the model, comparison, metric, and source location; rows must not be pooled into a cross-paper ranking. The diagnostic proposals in Table S2 are distinct from these reported results.

TABLE S4
Within-study evidence for objective, safety, and data-mixture trade-offs.

Source / contrast	Conditions and reported values	Supported inference	Boundary
mDPO [29], Table 1: DPO versus mDPO	Bunny-v1.0-3B, 10K preference examples: Object HalBench CHAIR _s 44.3 to 27.0; AMBER object coverage 74.1 to 67.4.	Fewer hallucinated descriptions accompany lower coverage on another benchmark under this setting.	Different benchmarks and metrics; this is not a universal grounding-utility trade-off or a measure of the Intent Gap.
mDPO [29], Table 2: objective ablation	Bunny-v1.0-3B: full mDPO CHAIR _s 27.0; without conditional preference 40.3; without reward anchor 34.3; without both 44.3.	Both components contribute in the reported ablation.	An output-level improvement does not isolate encoding, fusion, or decoding as its cause.
VLGuard [30], Appendix Table 14: language versus vision-language checkpoint	XSTest Unsafe, Llama-Guard judge: Vicuna-v1.5-7B attack success 1.50%; LLaVA-v1.5-7B 11.00%.	Safety of a language backbone need not survive the multimodal training pipeline.	Checkpoints differ in their training; this contrast does not isolate one architectural component or establish safety under every attack.
MM1 [31], Fig. 5a: caption/interleaved mixture	Caption-only versus 50:50 mixture: mean zero-shot 39.3 to 33.4; mean eight-shot 45.0 to 62.2.	The preferred mixture depends on the prompting regime.	Reported aggregate benchmarks, not direct causal-grounding scores; neither mixture is a universal optimum.
MM1 [31], Fig. 5b: adding text-only data	Ablation with 10% text-only data examines zero- and few-shot performance and TextCore separately.	Text replay and interleaved supervision serve different capability targets.	Preserve the other mixture components and task breakdown when interpreting transfer; overall accuracy can hide contrary task effects.

S6 SEARCH AND VERSION BOUNDARIES

The main survey states coverage through August 2026 with a selective update on 9 September 2026. This is a curated survey, not a preregistered systematic review: no retrospectively reconstructed screening counts or completeness guarantees are claimed. The companion catalog records 275 distinct papers cited in the main text, their section locations, and bibliographic metadata. The complete timeline contains a subset of 163 studies; the print-sized selection contains 70. Timeline omissions therefore do not mean exclusion from the surveyed literature. Publication-year buckets follow the cited version; earlier preprints can precede the displayed year.

S7 INTERPRETATION AND SCOPE BOUNDARIES

The contribution of the main survey is the joint synthesis of intervention mechanisms, supervision assumptions, diagnostic evidence, and application constraints, not the invention of every axis in the comparison matrix. The comparisons retain foundational vision–language models when they explain an MLLM mechanism and adjacent applications when they expose a transferable constraint. Core MLLM instruction or behavioral alignment is distinguished from image–text representation learning, multimodal entity alignment, and multimodal prediction without an LLM.

Graph augmentation is treated as an architectural mechanism, with application evidence used to explain its consequences. Recommendation and climate cases illustrate supervision and evidence constraints without claiming exhaustive domain coverage. GUI-agent control and multimodal content generation are outside comprehensive coverage. Text-only preference-learning studies are methodological background, not direct demonstrations of multimodal alignment.

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